

ME100 - ENGINEERING MECHANICS

ASSIGNMENT – 1

Rules for submission

Always proofread your written work. Check it using the following list of rules before you submit it.

1. Use A4 size paper ONLY.
2. Draw MARGINS on all sides.
3. Assignments must be HAND WRITTEN ONLY.
4. Handwriting should be NEAT AND READABLE.
5. Solve the problems STEP BY STEP.
6. Use either BLUE OR BLACK PEN OR BOTH.
7. STAPLE your pages together before handing them in (top left corner).
8. FRONT PAGE of the assignment should carry the following details ONLY.

FULL NAME:

ROLL NUMBER:

DEPARTMENT:

SECTION:

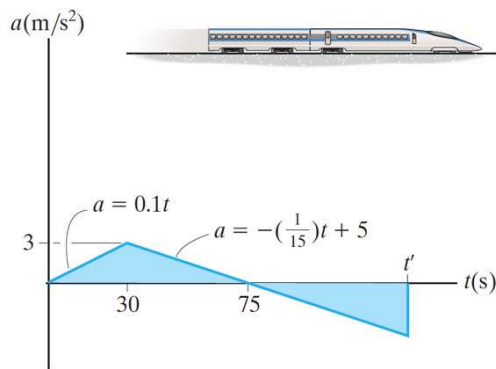
SUBJECT CODE AND NAME:

ASSIGNMENT NO:

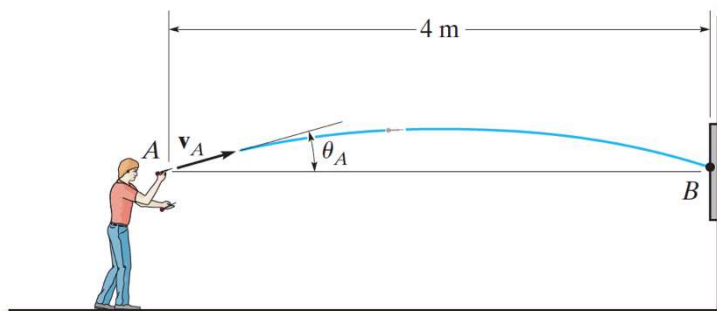
9. Submit the assignment on time.
10. Last date for submission: **ON or BEFORE 06th November 2024 10:00AM**
11. Late submission of assignments will not be accepted on any account.

QUESTIONS

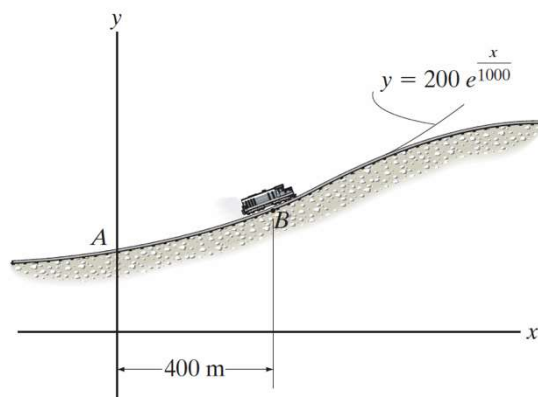
1. The a - t graph of the bullet train is shown. If the train starts from rest, determine the elapsed time before it again comes to rest. What is the total distance travelled during this time interval? Construct the v - t and s - t graphs.



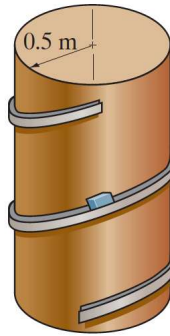
2. A rocket is fired from rest at $x = 0$ and travels along a parabolic trajectory described by $y^2 = [120(10^3)x]$ m. If the x component of acceleration is $a_x = (t^2/4)$ m/s², where t is in seconds, determine the magnitude of the rocket's velocity and acceleration when $t = 10$ s.
3. If the dart is thrown with a speed of 10 m/s, determine the shortest possible time before it strikes the target. Also, what is the corresponding angle θ_A at which it should be thrown, and what is the velocity of the dart when it strikes the target?



4. The train passes point A with a speed of 30 m/s and begins to decrease its speed at a constant rate of $a_t = -0.25$ m/s². Determine the magnitude of the acceleration of the train when it reaches point B, where $s_{AB} = 412$ m.



5. The box slides down the helical ramp such that $r = 0.5 \text{ m}$, $\theta = (0.5t^3) \text{ rad}$, and $z = (2 - 0.2t^2) \text{ m}$, where t is in seconds. Determine the magnitudes of the velocity and acceleration of the box at the instant $\theta = 2\pi \text{ rad}$.

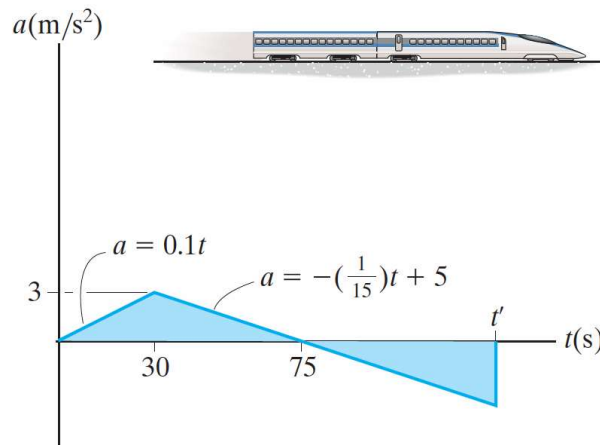


ME100 - ENGINEERING MECHANICS

ASSIGNMENT – 1

Solutions

1. The a - t graph of the bullet train is shown. If the train starts from rest, determine the elapsed time before it again comes to rest. What is the total distance travelled during this time interval? Construct the v - t and s - t graphs.



v - t Graph: For the time interval $0 \leq t < 30$ s, the initial condition is $v = 0$ when $t = 0$ s.

$$\begin{aligned} \left(\pm \right) \quad dv &= a dt \\ \int_0^v dv &= \int_0^t 0.1t dt \\ v &= (0.05t^2) \text{ m/s} \end{aligned}$$

When $t = 30$ s,

$$v|_{t=30 \text{ s}} = 0.05(30^2) = 45 \text{ m/s}$$

or the time interval $30 \text{ s} < t \leq t'$, the initial condition is $v = 45$ m/s at $t = 30$ s.

$$\begin{aligned} \left(\pm \right) \quad dv &= a dt \\ \int_{45 \text{ m/s}}^v dv &= \int_{30 \text{ s}}^t \left(-\frac{1}{15}t + 5 \right) dt \\ v &= \left(-\frac{1}{30}t^2 + 5t - 75 \right) \text{ m/s} \end{aligned}$$

Thus, when $v = 0$,

$$0 = -\frac{1}{30}t'^2 + 5t' - 75$$

Choosing the root $t' > 75$ s,

$$t' = 133.09 \text{ s} = 133 \text{ s}$$

Ans.

Also, the change in velocity is equal to the area under the a - t graph. Thus,

$$\begin{aligned}\Delta v &= \int a dt \\ 0 &= \frac{1}{2}(3)(75) + \frac{1}{2} \left[\left(-\frac{1}{15}t' + 5 \right) (t' - 75) \right] \\ 0 &= -\frac{1}{30}t'^2 + 5t' - 75\end{aligned}$$

This equation is the same as the one obtained previously.

The slope of the v - t graph is zero when $t = 75$ s, which is the instant $a = \frac{dv}{dt} = 0$. Thus,

$$v|_{t=75 \text{ s}} = -\frac{1}{30}(75^2) + 5(75) - 75 = 112.5 \text{ m/s}$$

The v - t graph is shown in Fig. *a*.

s - t Graph: Using the result of v , the equation of the s - t graph can be obtained by integrating the kinematic equation $ds = v dt$. For the time interval $0 \leq t < 30$ s, the initial condition $s = 0$ at $t = 0$ s will be used as the integration limit. Thus,

$$\begin{aligned}(\rightarrow) \quad ds &= v dt \\ \int_0^s ds &= \int_0^t 0.05t^2 dt \\ s &= \left(\frac{1}{60}t^3 \right) \text{ m}\end{aligned}$$

When $t = 30$ s,

$$s|_{t=30 \text{ s}} = \frac{1}{60}(30^3) = 450 \text{ m}$$

For the time interval $30 \text{ s} < t \leq t' = 133.09$ s, the initial condition is $s = 450$ m when $t = 30$ s.

$$\begin{aligned}(\rightarrow) \quad ds &= v dt \\ \int_{450 \text{ m}}^s ds &= \int_{30 \text{ s}}^{t'} \left(-\frac{1}{30}t^2 + 5t - 75 \right) dt \\ s &= \left(-\frac{1}{90}t^3 + \frac{5}{2}t^2 - 75t + 750 \right) \text{ m}\end{aligned}$$

When $t = 75$ s and $t' = 133.09$ s,

$$s|_{t=75 \text{ s}} = -\frac{1}{90}(75^3) + \frac{5}{2}(75^2) - 75(75) + 750 = 4500 \text{ m}$$

$$s|_{t=133.09 \text{ s}} = -\frac{1}{90}(133.09^3) + \frac{5}{2}(133.09^2) - 75(133.09) + 750 = 8857 \text{ m} \quad \text{Ans.}$$

The s - t graph is shown in Fig. *b*.

When $t = 30$ s,

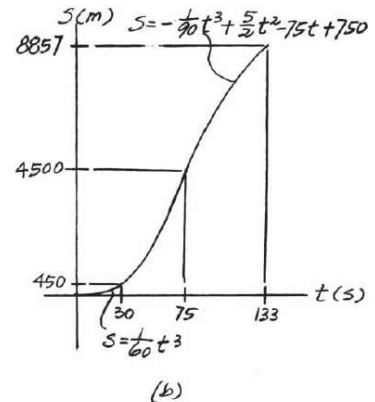
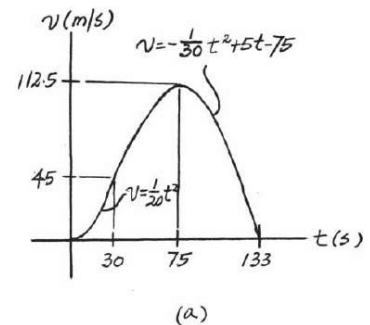
$$v = 45 \text{ m/s and } s = 450 \text{ m.}$$

When $t = 75$ s,

$$v = v_{\max} = 112.5 \text{ m/s and } s = 4500 \text{ m.}$$

When $t = 133$ s,

$$v = 0 \text{ and } s = 8857 \text{ m.}$$



2. A rocket is fired from rest at $x = 0$ and travels along a parabolic trajectory described by $y^2 = [120(10^3)x]$ m. If the x component of acceleration is $a_x = (t^2/4)$ m/s², where t is in seconds, determine the magnitude of the rocket's velocity and acceleration when $t = 10$ s.

Position: The parameter equation of x can be determined by integrating a_x twice with respect to t .

$$\begin{aligned}\int dv_x &= \int a_x dt \\ \int_0^{v_x} dv_x &= \int_0^t \frac{1}{4} t^2 dt \\ v_x &= \left(\frac{1}{12} t^3 \right) \text{ m/s} \\ \int dx &= \int v_x dt \\ \int_0^x dx &= \int_0^t \frac{1}{12} t^3 dt \\ x &= \left(\frac{1}{48} t^4 \right) \text{ m}\end{aligned}$$

Substituting the result of x into the equation of the path,

$$\begin{aligned}y^2 &= 120(10^3) \left(\frac{1}{48} t^4 \right) \\ y &= (50t^2) \text{ m}\end{aligned}$$

Velocity:

$$v_y = \dot{y} = \frac{d}{dt}(50t^2) = (100t) \text{ m/s}$$

When $t = 10$ s,

$$v_x = \frac{1}{12}(10^3) = 83.33 \text{ m/s} \quad v_y = 100(10) = 1000 \text{ m/s}$$

Thus, the magnitude of the rocket's velocity is

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{83.33^2 + 1000^2} = 1003 \text{ m/s} \quad \textbf{Ans.}$$

Acceleration:

$$a_y = \dot{v}_y = \frac{d}{dt}(100t) = 100 \text{ m/s}^2$$

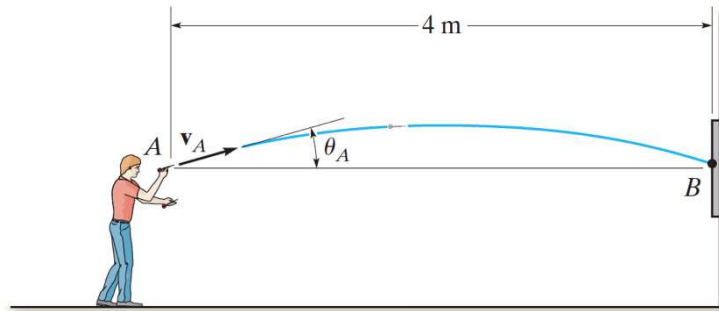
When $t = 10$ s,

$$a_x = \frac{1}{4}(10^2) = 25 \text{ m/s}^2$$

Thus, the magnitude of the rocket's acceleration is

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{25^2 + 100^2} = 103 \text{ m/s}^2 \quad \textbf{Ans.}$$

3. If the dart is thrown with a speed of 10 m/s, determine the shortest possible time before it strikes the target. Also, what is the corresponding angle θ_A at which it should be thrown, and what is the velocity of the dart when it strikes the target?



Coordinate System. The origin of the x - y coordinate system will be set to coincide with point A as shown in Fig. a .

Horizontal Motion. Here, $(v_A)_x = 10 \cos \theta_A \rightarrow$, $(s_A)_x = 0$ and $(s_B)_x = 4 \text{ m} \rightarrow$.

$$\begin{aligned} (+\rightarrow) \quad (s_B)_x &= (s_A)_x + (v_A)_x t \\ 4 &= 0 + 10 \cos \theta_A t \\ t &= \frac{4}{10 \cos \theta_A} \end{aligned} \quad (1)$$

Also,

$$(+\rightarrow) \quad (v_B)_x = (v_A)_x = 10 \cos \theta_A \rightarrow \quad (2)$$

Vertical Motion. Here, $(v_A)_y = 10 \sin \theta_A \uparrow$, $(s_A)_y = (s_B)_y = 0$ and $a_y = 9.81 \text{ m/s}^2 \downarrow$

$$\begin{aligned} (+\uparrow) \quad (s_B)_y &= (s_A)_y + (v_A)_y t + \frac{1}{2} a_y t^2 \\ 0 &= 0 + (10 \sin \theta_A) t + \frac{1}{2} (-9.81) t^2 \\ 4.905 t^2 - (10 \sin \theta_A) t &= 0 \\ t(4.905 t - 10 \sin \theta_A) &= 0 \end{aligned}$$

Since $t \neq 0$, then

$$4.905 t - 10 \sin \theta_A = 0 \quad (3)$$

Also

$$\begin{aligned} (+\uparrow) \quad (v_B)_y^2 &= (v_A)_y^2 + 2 a_y [(s_B)_y - (s_A)_y] \\ (v_B)_y^2 &= (10 \sin \theta_A)^2 + 2 (-9.81) (0 - 0) \\ (v_B)_y &= -10 \sin \theta_A = 10 \sin \theta_A \downarrow \end{aligned} \quad (4)$$

Substitute Eq. (1) into (3)

$$\begin{aligned} 4.905 \left(\frac{4}{10 \cos \theta_A} \right) - 10 \sin \theta_A &= 0 \\ 1.962 - 10 \sin \theta_A \cos \theta_A &= 0 \end{aligned}$$

Using the trigonometry identity $\sin 2\theta_A = 2 \sin \theta_A \cos \theta_A$, this equation becomes

$$1.962 - 5 \sin 2\theta_A = 0$$

$$\sin 2\theta_A = 0.3924$$

$$2\theta_A = 23.10^\circ \text{ and } 2\theta_A = 156.90^\circ$$

$$\theta_A = 11.55^\circ \text{ and } \theta_A = 78.45^\circ$$

Since the shorter time is required, Eq. (1) indicates that smaller θ_A must be chosen. Thus

$$\theta_A = 11.55^\circ = 11.6^\circ \quad \text{Ans.}$$

and

$$t = \frac{4}{10 \cos 11.55^\circ} = 0.4083 \text{ s} = 0.408 \text{ s} \quad \text{Ans.}$$

Substitute the result of θ_A into Eq. (2) and (4)

$$(v_B)_x = 10 \cos 11.55^\circ = 9.7974 \text{ m/s} \rightarrow$$

$$(v_B)_y = 10 \sin 11.55^\circ = 2.0026 \text{ m/s} \downarrow$$

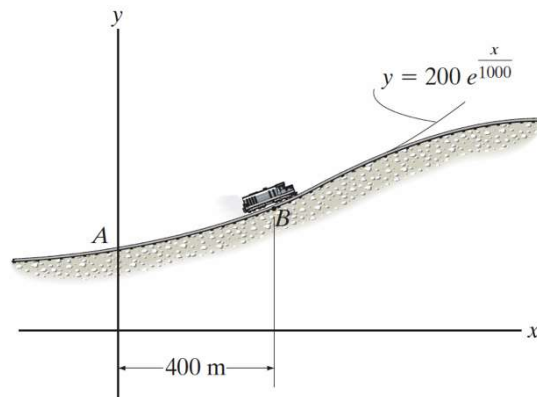
Thus, the magnitude of v_B is

$$v_B = \sqrt{(v_B)_x^2 + (v_B)_y^2} = \sqrt{9.7974^2 + 2.0026^2} = 10 \text{ m/s} \quad \text{Ans.}$$

And its direction is defined by

$$\theta_B = \tan^{-1} \left[\frac{(v_B)_y}{(v_B)_x} \right] = \tan^{-1} \left(\frac{2.0026}{9.7974} \right) = 11.55^\circ = 11.6^\circ \quad \nwarrow \quad \text{Ans.}$$

4. The train passes point A with a speed of 30 m/s and begins to decrease its speed at a constant rate of $a_t = -0.25 \text{ m/s}^2$. Determine the magnitude of the acceleration of the train when it reaches point B, where $s_{AB} = 412 \text{ m}$.



Velocity: The speed of the train at B can be determined from

$$v_B^2 = v_A^2 + 2a_t(s_B - s_A)$$

$$v_B^2 = 30^2 + 2(-0.25)(412 - 0)$$

$$v_B = 26.34 \text{ m/s}$$

Radius of Curvature:

$$y = 200e^{\frac{x}{1000}}$$

$$\frac{dy}{dx} = 0.2e^{\frac{x}{1000}}$$

$$\frac{d^2y}{dx^2} = 0.2(10^{-3})e^{\frac{x}{1000}}$$

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{d^2y}{dx^2}\right|} = \frac{\left[1 + \left(0.2e^{\frac{x}{1000}}\right)^2\right]^{3/2}}{\left|0.2(10^{-3})e^{\frac{x}{1000}}\right|} \bigg|_{x=400 \text{ m}} = 3808.96 \text{ m}$$

Acceleration:

$$a_t = \dot{v} = -0.25 \text{ m/s}^2$$

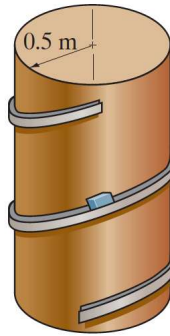
$$a_n = \frac{v^2}{\rho} = \frac{26.34^2}{3808.96} = 0.1822 \text{ m/s}^2$$

The magnitude of the train's acceleration at B is

$$a = \sqrt{a_t^2 + a_n^2} = \sqrt{(-0.25)^2 + 0.1822^2} = 0.309 \text{ m/s}^2$$

Ans.

5. The box slides down the helical ramp such that $r = 0.5$ m, $\theta = (0.5t^3)$ rad, and $z = (2 - 0.2t^2)$ m, where t is in seconds. Determine the magnitudes of the velocity and acceleration of the box at the instant $\theta = 2\pi$ rad.



Time Derivatives:

$$r = 0.5 \text{ m}$$

$$\dot{r} = \ddot{r} = 0$$

$$\dot{\theta} = (1.5t^2) \text{ rad/s}$$

$$\ddot{\theta} = (3t) \text{ rad/s}^2$$

$$z = 2 - 0.2t^2$$

$$\dot{z} = (-0.4t) \text{ m/s}$$

$$\ddot{z} = -0.4 \text{ m/s}^2$$

When $\theta = 2\pi$ rad,

$$2\pi = 0.5t^3$$

$$t = 2.325 \text{ s}$$

Thus,

$$\dot{\theta}|_{t=2.325 \text{ s}} = 1.5(2.325)^2 = 8.108 \text{ rad/s}$$

$$\ddot{\theta}|_{t=2.325 \text{ s}} = 3(2.325) = 6.975 \text{ rad/s}^2$$

$$\dot{z}|_{t=2.325 \text{ s}} = -0.4(2.325) = -0.92996 \text{ m/s}$$

$$\ddot{z}|_{t=2.325 \text{ s}} = -0.4 \text{ m/s}^2$$

Velocity:

$$v_r = \dot{r} = 0$$

$$v_\theta = r\dot{\theta} = 0.5(8.108) = 4.05385 \text{ m/s}$$

$$v_z = \dot{z} = -0.92996 \text{ m/s}$$

Thus, the magnitude of the box's velocity is

$$v = \sqrt{v_r^2 + v_\theta^2 + v_z^2} = \sqrt{0^2 + 4.05385^2 + (-0.92996)^2} = 4.16 \text{ m/s} \quad \text{Ans.}$$

Acceleration:

$$a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 0.5(8.108)^2 = -32.867 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0.5(6.975) + 2(0)(8.108)^2 = 3.487 \text{ m/s}^2$$

$$a_z = \ddot{z} = -0.4 \text{ m/s}^2$$

Thus, the magnitude of the box's acceleration is

$$a = \sqrt{a_r^2 + a_\theta^2 + a_z^2} = \sqrt{(-32.867)^2 + 3.487^2 + (-0.4)^2} = 33.1 \text{ m/s}^2 \quad \mathbf{Ans.}$$